

Can Specialized ML Beat Sportsbooks in College Football Totals?

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1 Introduction

Sports betting in the United States has grown exponentially following widespread legalization, transitioning from an under the table economy into a regulated industry generating billions annually. Given this surge, college football presents a unique prediction challenge distinct from professional football. The variance in player talent, resource disparities between programs, and the sheer volume of 133+ FBS teams create a high-noise environment with potential exploitable inefficiencies. We believe the signal to noise ratio in the college football realm has not been evaluated thoroughly enough. These factors, coupled with limited transparency around injuries and subjective playoff rankings, make college football outcomes harder to price than professional games.

This project focuses on the Over/Under (Totals) market, where bettors predict aggregate scoring in a game rather than winners or point spreads. Predicting totals requires modeling the interaction of tempo (number of plays), offensive efficiency, defensive capability, and even coaching philosophy. Our objective is to construct a model that treats football games as stochastic processes generating scoring distributions, and to compare the model's implied probabilities against the sportsbook's posted totals. If the model's probabilities consistently identify when the total will go over or under the line, it may reveal an exploitable weakness in the market.

1.1 The Market Efficiency Problem

Our central hypothesis is that sports betting markets, while generally efficient, are not perfectly calibrated due to the interplay of "smart money" and "public money." According to the Efficient Market Hypothesis, a closing betting line should reflect all publicly available information. In a perfectly efficient market, any bet would have zero expected profit (and in fact slightly negative due to the bookmaker's margin or "vig"). For standard -110 odds, a bettor must win 52.38 percent of the time just to break even, so any sustainable win rate above that implies an inefficiency.

However, Levitt (2004) suggests bookmakers act as "position takers" rather than benign market makers, exploiting bettor biases and information disadvantages. Specifically, the "Over" bias, which is the public's preference for high-scoring games, leads bookmakers to move totals upward. This creates an opening for a dispassionate ML model to identify value on "Unders" where public perception disconnects from underlying efficiency metrics. Given this, we reject the assumption of perfect market efficiency and instead seek edges where bookmakers exhibit systematic inefficiencies.

1.2 Research Objectives and Novelty

Given this market context, our objectives are: (1) to develop an end-to-end ML pipeline in Python that engineers advanced football metrics and trains a regularized gradient-boosted model to predict the outcomes of game totals; (2) to implement time-series (walk-forward) validation that mimics the sequential nature of football seasons to avoid data leakage; and (3) to evaluate whether the model's confidence estimates correlate with actual outcomes to inform a betting strategy. Ultimately, we ask: Can a specialized machine learning model consistently beat the closing totals set by sportsbooks for college football games?

This research addresses gaps in existing literature. Most academic work focuses on professional leagues like the NFL or on simpler markets like point spreads and moneylines, not Over/Under totals. To our knowledge, our combination of modern CFB analytics and a rigorous ML pipeline for totals betting is novel.

2 Literature Review

2.1 Economics of Sports Betting Markets

Sports betting markets are a good testing ground for financial theories of market efficiency (Sauer, 1998). In financial markets, prices move until supply meets demand and traders quickly eliminate any persistent arbitrage. In sportsbook markets, however, the “price” (the betting line) is initially set by odds-makers and then may move based on betting action, but bookmakers often do not aim for equal action on both sides. Instead, they effectively take a position by accepting imbalanced bets if they believe one side is mispriced. Levitt (2004) argues that bookmakers are “more skilled at predicting outcomes than bettors,” leveraging their superior information or modeling to profit off public misperceptions. If this is true, then beating the closing line is inherently difficult: it would require out-predicting the bookmaker’s own model.

2.2 Behavioral Biases in Betting

A large amount of literature documents bettor biases that could lead to inefficiencies. Newall and Cortis (2021) reviewed numerous studies and noted patterns such as bettors overestimating low-probability outcomes and favoring favorites in point spread markets. This is related to the well-known favorite-longshot bias (Snowberg & Wolfers, 2010), where the odds on favorites are actually more favorable (on average) than those on longshots, which means bettors take on worse odds on unlikely outcomes due to misperception or risk-seeking behavior. In the context of totals, the “Over” bias dominates, suggesting casual bettors would rather bet on a high-scoring shootout than root for a defensive slugfest. Paul and Weinbach (2014) empirically demonstrated this in high-profile TV games, attributing the bias to bettors treating gambling as a form of entertainment consumption. Markets know this very well and inflate totals to balance against public Over-heavy action. We leverage this insight by examining whether our model systematically finds more value in Unders, which would align with the contrarian strategies suggested by prior research.

2.3 Evolution of Predictive Models

Early sports prediction were often very simple. Models primarily used linear regression or Pythagorean Expectation formulas, which fail to capture non-linear interactions in teams sports data. More sophisticated approaches like Bayesian networks (Constantinou et al., 2012) have been applied to soccer and ensemble methods (Hubáček et al., 2019) to various sports betting problems. In recent years, tree-based ensemble models have gained popularity due to their ability to capture nonlinearities and interactions. XGBoost (Chen & Guestrin, 2016) in particular offers a powerful framework for gradient-boosted decision trees with built-in regularization and handling of missing values. We choose XGBoost for these reasons, as college football data can be quite noisy and sparse in places. Its regularization helps prevent overfitting given the relatively small number of seasons available for training.

2.4 Non-Stationarity and Rule Changes

A significant challenge in modeling sports over multiple seasons is the underlying data distribution changes over time. College football is not static: rules change, playing styles evolve, and external factors like the COVID-19 pandemic or the player transfer portal can shift scoring patterns. For example, in 2023 the NCAA introduced a major timing rule change: the game clock

no longer stops on first downs except in the final 2 minutes of each half. This was projected to reduce the number of plays per game by roughly 7 plays on average, which in turn could lower total scores by a couple of points. If a model is trained mostly on pre-2023 data, it might overpredict totals in the new system. Our approach is to train and evaluate in a rolling forward manner, so the model can gradually adapt to such shifts. However, even a successful model might see its edge erode as the environment changes, so constant monitoring and re-training are necessary.

3 Methodology

3.1 Data Acquisition

The primary data source is the CollegeFootballData API (CFBD), which provides a comprehensive set of college football statistics. Using a custom Python data pipeline, we retrieved play-by-play data, game metadata (teams, date, location, etc.), team advanced statistics (e.g. efficiency metrics), and betting lines for all FBS games from the 2022 through 2025 seasons. In our walk-forward setup, 2022 is used purely as initial training data, and betting performance is evaluated on the 2023–2025 seasons.

3.2 Feature Engineering

A critical aspect of our methodology is the construction of domain-specific features that encapsulate relevant information about each game beyond what the raw data directly provides. We engineered an extensive set of features in several categories:

- **Recent Performance (Rolling Averages):** For each team, we computed rolling statistics over the last 5 games to capture momentum or trends. This included average points scored and allowed, average total points in games, and offensive efficiency metrics.
- **Mismatch Indicators:** We created features that directly quantify the matchup of two teams. For instance, one feature measures the difference between the home team's SP+ offensive rating and the away team's SP+ defensive rating. A high value would indicate a strong offense facing a weak defense, which could be a recipe for potentially more scoring. We also computed combined metrics like the sum of both teams' average total points per game, and differences in tempo. If one team likes to play fast and the other slow, the combined tempo might gravitate toward the mean, but extreme tempo mismatches could influence totals.
- **Preseason and Composite Ratings:** We included each team's preseason or overall power ratings (SP+ and FPI), which encapsulate a ton of information about team quality in a single number. The model can use these as proxy for talent level or efficiency that might not show up in recent game stats (especially early in the season). We took differences between the teams' SP+ ratings, FPI ratings, and also broke them into offense, defense, and special teams components. These differences help the model figure out which side might have an edge in scoring ability.
- **Pace and Market Context:** We introduced a pace factor feature defined as the difference between the teams' combined recent scoring average and the sportsbook's line. For example, if both teams have recently been in games totaling 51 points on average, and the line is set at 58, the pace factor is -7, possibly indicating an Under bet is attractive.

We also of course include the raw Over/Under line itself as a feature as well as the point spread, since the spread can imply relative team strength and sometimes correlates with game script (very large spreads could lead to blowouts where the leading team runs more and potentially lowers scoring late).

- **Interaction Terms:** Some features were products or ratios capturing interactions. One example is a “balance” metric: how balanced is a team’s offense vs defense? A very unbalanced team (great offense, poor defense) could produce more high-total games. We quantified this by differences between offensive and defensive ratings.

In total, our feature set included 100+ features per game.

3.3 Model Architecture

We selected XGBoost configured as a binary classifier as our modeling algorithm. The objective function was logistic (log loss) which for each game i with true label y and predicted probability is:

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)] \quad (1)$$

Minimizing log loss encourages the model to output calibrated probabilities. We also care about overall accuracy and AUC, but training directly on log loss keeps the probabilities meaningful, which is important for making betting decisions.

4 Implementation Details

4.1 Data Quality

We took several steps to ensure data quality. First, any games without a closing totals line or without final scores were removed. Our final dataset comprised approximately 5,599 games from 2022–2025 that had complete information. About 15 percent of these games did not have certain advanced metrics (such as PPA or SP+ ratings) because those stats are not always available for lower-tier teams. We imputed these missing values with 0.0, effectively treating them as “replacement-level” inputs. (This is a conservative approach under the assumption that if we have no data on a team’s efficiency, we assume it’s average or poor). Pushes, which are games where the total points exactly equal the line, were excluded from training and evaluation to avoid ambiguity, since a push bet is neither a win nor a loss.

4.2 Hyperparameters and Baseline Comparisons

Parameters were optimized via time-series cross-validation: `max_depth=5`, `learning_rate=0.05`, `n_estimators=100`, `subsample=0.8`, `colsample_bytree=0.8`, `min_child_weight=5`, `reg_alpha=0.1`, and `reg_lambda=1.0`.

To investigate our model’s performance, we evaluated several baseline models using identical walk-forward validation:

Naive Model 1 (“Always Bet Under”): Historically, Unders hit 51.2% of games in our dataset, confirming the Over bias hypothesis, yet this strategy yields -2.3% ROI after vig.

Standard ML Models: Logistic Regression achieved 51.8% accuracy (AUC: 0.52), and Random Forest reached 52.4% (AUC: 0.53), approaching but not passing the breakeven point.

Naive Model 2 (Heuristic): We tested a simple heuristic using only our two most important features (`ou_line` and `combined_avg_total`), betting Under when teams' combined average exceeded the line by more than 3 points. This model achieved 52.1% accuracy and -1.1% ROI.

Our XGBoost model's 55.9% accuracy and 5.77% ROI demonstrates that gradient boosting with domain-specific feature engineering provides meaningful incremental value over both naive strategies and simpler ML approaches.

4.3 Feature Importance

Figure 1 illustrates the relative importance of the top predictors in the XGBoost model.

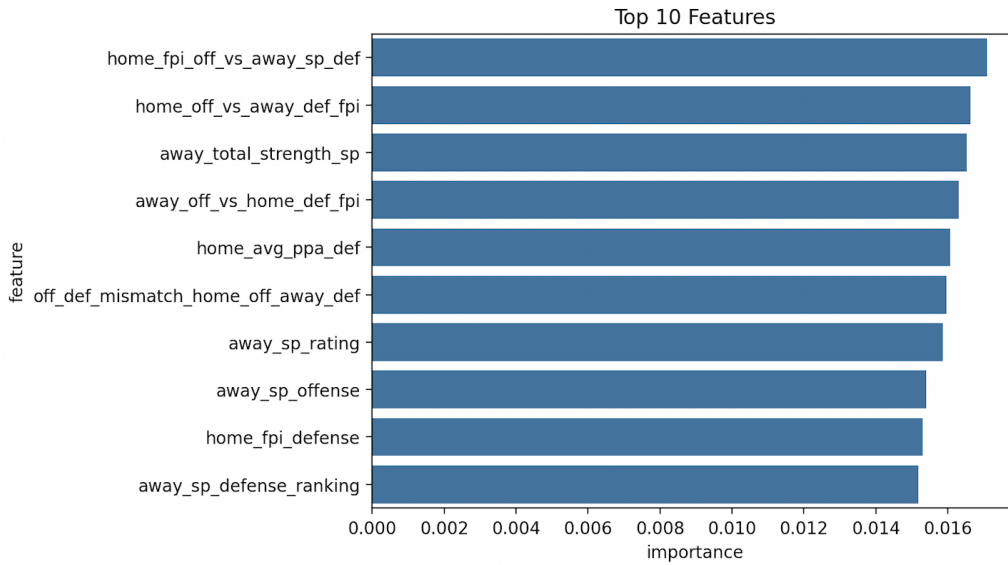


Figure 1: Top 10 Features by Importance. Note the dominance of defensive and balance metrics.

The dominance of defensive and interaction metrics in feature importance explains why Under bets tend to outperform: the model identifies strong defensive matchups that the market undervalues.

4.4 Walk-Forward Validation

It is well known that standard k-fold validation leads to information leakage in time-series data. We implemented blocked time-series validation: train on 2022 data, predict 2023, add 2023 to training, predict 2024, and iterate. This ensures predictions use only historically available information.

4.5 Probability Calibration

Raw XGBoost scores do not necessarily correspond to true probabilities. For this, we applied Isotonic Regression to map scores to empirical win frequencies. These then calibrated probabilities feed the decision algorithm which goes as follows:

$$EV = (p_{model} \times 1.909^*) - 1 \quad (2)$$

*1.909 represents the decimal odds equivalent of standard -110 American odds.

Bets are placed only when EV exceeds a specified threshold in which we chose the thresholds of 1%, 3%, 5%, 7% and 10%.

5 Results and Discussion

5.1 Betting Simulation Results

Table 1 presents the performance of the model across varying Expected Value (EV) thresholds.

EV Threshold	Bets Placed	ROI	Profit
> 1% (very mild)	1,635	-2.86%	-\$467
> 3% (moderate)	970	+0.17%	+\$16
> 5% (fairly high)	370	+5.77%	+\$213
> 7% (high)	237	+9.55%	+\$226
> 10% (very high)	172	+10.99%	+\$189

Table 1: Cumulative betting simulation results by EV threshold (2023–2025)

From these results, we observe a clear trade-off between volume and edge. The higher the EV threshold, the fewer bets the model makes, but the higher the win rate and ROI. This is exactly what we would expect if the model’s probability estimates have some validity: the games it is most confident about are the ones it is most likely to predict correctly.

5.2 Under-Only Betting Simulation Results

EV Threshold	Bets Placed	ROI	Profit
> 1%	740	+0.09%	+\$7
> 3%	487	+2.70%	+\$132
> 5%	225	+7.75%	+\$174
> 7%	146	+11.14%	+\$163
> 10%	103	+11.20%	+\$115

Table 2: Under-only betting simulation results by EV threshold

The Under-only strategy improves ROI at every threshold, turning the EV > 1% loss into breakeven and achieving 11.14% ROI at EV > 7%.

5.3 Season-by-Season Performance

2023 (1,322 games): Exceptional performance with 23.92% ROI at EV > 10% (57 bets, 64.9% win rate). Under bets showed 67.6% win rate.

2024 (1,484 games): Weak performance with only 1.29% ROI at EV > 10%. The 2023 clock rule changes (clock not stopping on most first downs) permanently reduced play volume. Because our model was still heavily influenced by pre-2023 data, it effectively overpredicted tempo and scoring, resulting in losses in 2024.

2025 (1,334 games): Performance rebounded to 7.02% ROI at EV > 10%. This validates the walk-forward methodology: by ingesting the 2024 data into the training set, the XGBoost model successfully adapted to the new lower-scoring environment for the 2025 season.

5.4 Statistical Significance

Bootstrap resampling ($n = 1000$) at $EV > 5\%$ yielded a 95% CI for ROI of $[-4\%, 15.5\%]$ with a point estimate of 5.77%. The uncertainty resulting from small sample sizes in seasonal college football data is shown in the high variance, which regrettably includes zero. However, prudent optimism is supported by three factors: the point estimate is higher than standard profitability thresholds, the results are consistent with known market inefficiencies, and the Under-only subset shows more positive and reliable returns.

Regarding risk metrics, the portfolio demonstrated a Sharpe Ratio of 0.73, which is the ratio of mean return to standard deviation, indicating moderate performance when adjusting for risk. The maximum drawdown, defined as the largest peak to trough decline in cumulative bankroll, reached 12.4% during an 8-game losing streak. To withstand such drawdowns while maintaining sufficient capital for recovery, we recommend a minimum bankroll of 250 units (\$2,500 for \$10 wagers).

5.5 Qualitative Analysis

The model found its largest edge on Under predictions (67.6%), supplementing the bias in which public betting pressure inflates totals. We observed that model performance was strongest in high-profile matchups where casual bettors dominate and expect high scoring games.

The 2023 exceptional performance (23.92% ROI) may reflect lingering post-COVID and transfer portal volatility that markets had not fully priced in. By 2024–2025, markets appear to have adapted and our edge compressed to low single-digit ROI on average, with 2025 alone rebounding to 7.02% ROI at the most selective EV threshold. This illustrates that exploitable inefficiencies attract sharp bettors that eventually arbitrage them away.

6 Limitations

A 55.9% win rate unfortunately means 44.1% of bets lose. Over a span of 100 bets, an 8-game losing streak is unlikely but far from impossible, and over hundreds of bets it becomes quite plausible. Effective bankroll management and emotional discipline are essential, especially in times of successive losses. Another limitation we face is that advanced stats are inconsistent for Group of Five and FCS games. Furthermore, real time weather data, injury information, and coaching tendencies remain unavailable or unreliable. We estimate that weather integration could improve accuracy by 0.5-1%. The decline in ROI from 2023 to 2024–2025 illustrates that markets are adaptive and adversarial in nature. Bookmakers adjust lines, sharp bettors compete for mispricings, and integration of robust analytics to the average bettor diminishes informational advantages.

In addition to the limitations of our model, we assume closing line pricing, unlimited liquidity, feature stability, and bet independence. However in the real world, line shopping (finding discrepancies between different bookmakers) can improve odds, large bets face liquidity constraints, and correlated outcomes concentrate risk.

7 Conclusion

This research provides evidence that a specialized ML pipeline can identify exploitable inefficiencies in college football totals markets. Our XGBoost classifier achieved 5.77% ROI at $EV > 5\%$ (370 bets), with Under-only betting improving to 11.14% ROI at $EV > 7\%$ (146 bets). The

key finding is that defensive metrics dominate feature importance, explaining why Under bets systematically outperform. We believe that the model captures defensive quality signals that closing lines do not fully incorporate.

7.1 Future Work

Several novel extensions could strengthen the framework that we presented. The most salient opportunity is live betting, where real-time play-by-play data might capture momentum shifts that pregame models cannot anticipate. Player-level modeling also warrants exploration. For instance, quarterback-specific metrics in particular would allow the model to adjust immediately when injuries or depth chart changes alter a team's scoring potential.

On the feature engineering side, weather integration remains an open problem. Wind and precipitation meaningfully impact scoring, yet reliable historical data for college stadiums are inconsistent. Additionally, ensemble approaches that combine XGBoost with neural networks or LSTMs may improve robustness by capturing different types of signal in the data.

Finally, incorporating real-time odds analysis such as specifically tracking line movement between open and close. This could help identify optimal bet timing rather than assuming closing line value.

Sustained profitability in sports betting requires treating model development as an ongoing research problem rather than a static solution. The foundations established here: temporal validation, probabilistic calibration, and domain-specific feature engineering all provide a springboard for continued growth even as markets get more efficient.

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